Mapping Lithium-Bearing Pegmatite With Short-Wave Infrared (SWIR) Hyperspectral Imaging of Jingerquan Li–Be–Nb–Ta Pegmatite Deposit, Eastern Tianshan, NW China

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***Abstract*— Lithium (Li) is growing in importance and demand in several industrial applications, such as portable electric devices, electric vehicles, and hybrid electric vehicles. Li-rich pegmatites are one of the main sources of Li production in the world, so low-cost and high-efficiency exploration using remote sensing has become an important means for promoting the discovery of Li resources. Although imaging spectroscopy has great potential for Li-rich pegmatites identification and regional delineation, due to the limitations of resolution and data acquisition, the exploration of the Earth's surface at various scales on a variety of platforms requires further research. The purpose of this study is to use a ground-based HySpex imaging hyperspectrometer to map pegmatite zones directly, developing a new approach for mineralized pegmatite exploration. To achieve this, we present the minimum wavelength mapper (MWM), which combines the position and depth information of the deepest absorption feature, to give a per-pixel overview map of the Jingerquan Li–Be–Nb–Ta pegmatite deposit pit and heap. The results show that images between 2100 and 2450 nm provide useful information about mineral assemblages that display strong spectral features, such as Li-rich pegmatites, Li-poor pegmatites and alteration, and their spatial distribution at the surface in areas covered by the imagery. The wavelength position and the depth of Al–OH can provide an overview of the mineral assemblages and abundances, to rapidly classify possible regions of interest for further analysis.**

***Index Terms*— Hyperspectral imaging, HySpex, lithium (Li)-bearing pegmatite, minimum wavelength mapper (MWM).**

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I. Introduction

As a KEY metal element with strong chemical activity, lithium (Li) is widely used in aerospace, materials, metallurgy, light industry, chemicals, and other fields, especially in the green energy industry led by new vehicles using Li-batteries, which will lead to an unprecedented increase in the demand for Li [1], [2], [3]. The demand for Li is expected to increase by 20% every year, and the price of Li will also rise significantly [4], [5]. Li mainly occurs in Li-bearing pegmatite deposits and brines. More than half of the world's Li production comes from Li-bearing pegmatites [6], [7]. The main Li-bearing minerals include spodumene, montebrasite, petalite, and lepidolite. When compared with traditional exploration methods, optical remote sensing has the potential for fast-tracking cost-effective Li exploration.

Remote sensing supports lithological, structural, and hydrothermal alteration mapping [8], [9], which is reflected in its successful application in exploration for porphyry copper and gold deposits [10], [11], [12], as well as volcanogenic massive sulfide (VMS) ore deposits [13], [14], rare earth element (REE) exploration, and brine and evaporite deposits [15], [16], [17], [18], [19], [20]. Similarly, the application of remote sensing to Li mineralization began in the late 1980s. Lefevre [21] used band ratios for Landsat multispectral scanner system (MSS) images to detect geobotanical anomalies and concluded that spectral anomalies were related to the dispersal halos of metals, such as Li, tungsten, and arsenic. Later, researchers focused on the theme of remote sensing mapping Li-pegmatite with multispectral remote sensing data (ASTER, Landsat-5, Landsat-8, and Sentinel-2) and different image processing techniques. Mendes et al. [22] applied a spectral angle mapper

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(SAM) and mixture-tuned matched filtering (MTMF) methods to ASTER imagery for mapping Li-bearing pegmatites in the Vale do Jequitinhonha region, Brazil. Although both methods can correctly identify the spectral characteristics of pegmatite intrusion, they also generate some false-positives and misclassified pixels. Recently, Cardoso-Fernandes et al. proposed the application of RGB (red, green, and blue) combination, band ratio, and principal component analysis (PCA) in Sentinel-2 images to identify Li pegmatites based on the recognition of the associated alteration halos. Afterward, RGB and PCA were applied in different multispectral products, such as Landsat-5 thematic mapper (TM), Landsat-8 operational land imager (OLI) and thermal infrared sensor (TIRS), ASTER, and Sentinel-2 MSI, to directly identify hydrothermal alteration minerals and Li minerals [23], [24], [25], [26], [27], [28]. Despite the success in Li pegmatite discrimination, all these methods manifested spectral confusion with other minerals. Laukamp et al. [29] applied a multiple feature extraction method to airborne hyperspectral imagery, successfully discriminating pegmatites from other lithologies in the Eastern Fold Belt of the Mount Isa Inlier (Queensland). At the same time, because of the limitation of the relatively small outcrop size of Li-bearing pegmatites and the resolution of satellite images or airborne imaging, Li exploration needs to be further improved.

At present, most remote sensing studies have focused on the pegmatite identification and mapping of Li-bearing pegmatites using satellite multispectral data, and there is a lack of research on Li exploration using hyperspectral data [30]. Ground-based and drone-borne remote sensing technology can improve the spatial resolution of imaging and provide an effective visualization means for mining exploration. Boesche et al. [31] and Krupnik and Khan [32] obtained ground-based, visible and near-infrared (VNIR) and shortwave infrared (SWIR) hyperspectral data to map the neodymium distribution on the surface of carbonate in outcrops. Recently, several studies combined ground-based hyperspectral images with 3-D surface models, integrating a 3-D digital model and 3-D photogrammetric data to finely visualize the hyperspectral data [33], [34], [35]. However, because of the technological limitations of currently available hyperspectral sensors and data processing, most of the spectral coverage of hyperspectral sensors carried by lightweight unmanned aerial vehicles (UAVs) and ground-based imaging is in the VNIR range [36], [37], [38]. In particular, Booysen et al. [39] used a multiscale hyperspectral imaging approach to map Li-bearing minerals and their spatial distribution and demonstrated that Li-bearing minerals in outcrops can be directly mapped using hyperspectral imaging. Outcrop sensing is still impeded by factors, such as the crystal sizes and the mineral complexity of pegmatites. Notably, the Tschermark exchange of mica series in pegmatites can lead to regular changes in Al–OH absorption features. The specific analysis is as follows: the crystal structure shows that the Si–O tetrahedron and Al–O octahedron are connected in a T–O–T manner, and its formula can hence be written as follows:

$$XY_{\sim 2}\square_{1-0}Z_4O_{10}(W)_2 \tag{1}$$

where X is the interlayer cation, mainly K^+ and Na^+; Y is an octahedral cation, mainly $^{VI}Al^{3+}$, Fe^{2+}, and Mg^{2+}; $\square$ is a vacancy; Z is a tetrahedral cation, mainly Si^{4+} and $^{IV}Al^{3+}$; and W is an anion, mainly OH^- and F^-.

In single-scale data sources, the extracted absorption features of the reflectance spectra collected from outcrop sensing can be quickly extracted to generate different lithology, alteration, and other content distribution maps, which provide important information for geologists to evaluate a mining area in a cost and time-efficient manner. Therefore, it is important to extract the distribution information of rocks and minerals according to the shape parameters of spectral features, such as the position, depth, width, area, and asymmetry in hyperspectral data. Minimum wavelength mapper (MWM) [40] combines position and depth information of the deepest absorption feature to provide a pixel-by-pixel overview of dominating mineral occurrences. This method has the characteristics of unsupervised and repeatable classification, which can provide an effective means for direct mapping in hyperspectral images. This method has also been successfully applied to hyperspectral drill core data by combining the 1st, 2nd, and 3rd deepest absorptions in a given wavelength range [41].

This study aims to explore the imaging spectral characteristics of different mineral assemblages in the Jingerquan Li–Be–Nb–Ta pegmatite deposit and interprets the spatial distribution and deposit genesis from the perspective of remote sensing, laying the foundation for subsequent pegmatite research and the exploration of Li. We evaluate the HySpex hyperspectral imager to map Li-rich pegmatites (such as the spodumene-bearing pegmatite zone, lepidolite-bearing pegmatite zone, and Li-mica-spodumene pegmatites), Li-poor pegmatites (nonmineralized pegmatite zone, quartz–feldspar zone, muscovite zone, and so on), and alteration directly in the mining outcrop area to provide a novel method for the remote sensing-based characterization of Li-bearing pegmatites. MWM is used to calculate the wavelength position and depth of the deepest absorption feature between ~2100 and ~2450 nm, producing classification maps of an assessment of the dominant groups of minerals and their spatial distribution in the ore heap and pits. Because most of the SWIR active minerals found in the pegmatite contain an Al–OH feature, the deepest absorption feature wavelength position and depth image of hyperspectral images between ~2195 and ~2220 nm are used to map the occurrence of Al–OH, directly describing the spatial distribution and abundance map of the pegmatite zone and indicating the change in the absorption position of Li-rich pegmatites and Li-poor pegmatites. MWM-based maps are validated by collecting VNIR-SWIR reflectance spectra of selected samples in the laboratory using a FieldSpec4 high-resolution analytical spectral device (ASD).

II. Study Area

The East Tianshan orogenic belt is an important part of the Central Asian Orogenic Belt, which mainly includes the Halierke belt, the Jueluotage belt, and the Middle Tianshan massif [see Fig. 1(a)] [42]. The main rock types comprise late

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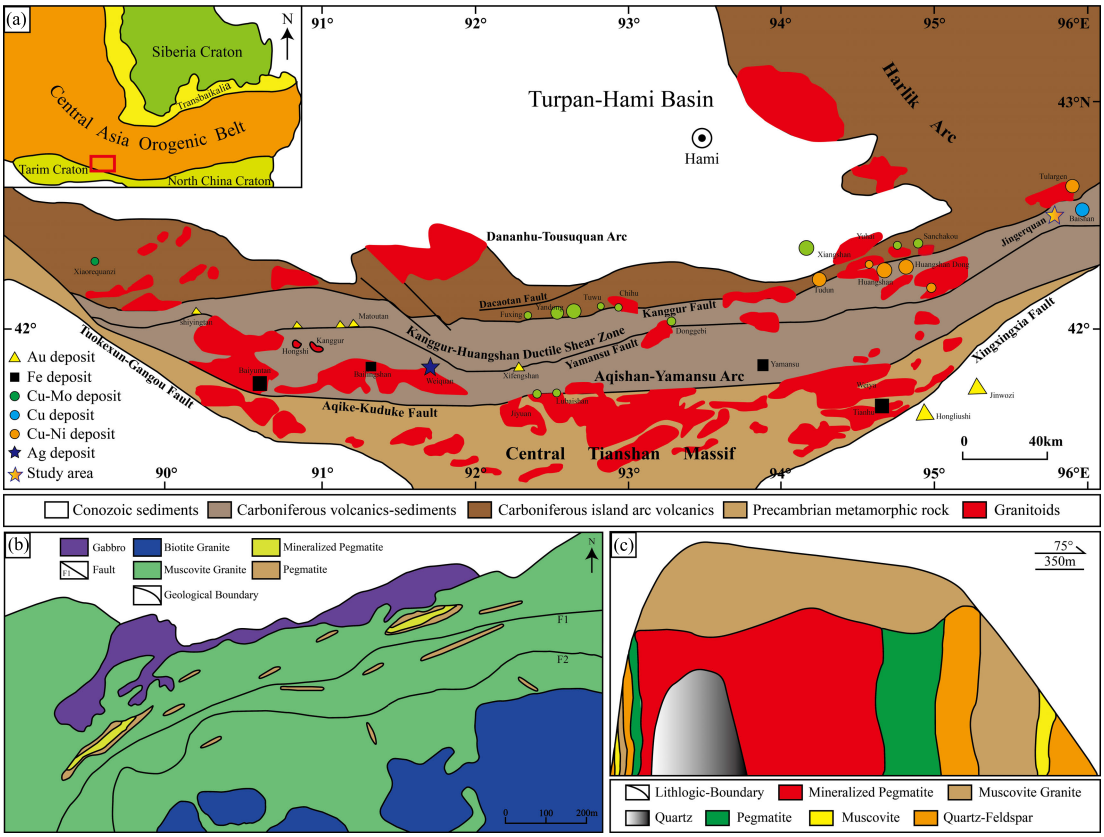


Fig. 1. (a) Regional geology and mineral distribution map of east Tianshan orogenic belt (modified from [51] and [52]). (b) Deposit geology of Jingerquan Li–Be–Nb–Ta mineralized pegmatite showing sampling locations (modified from [48]). (c) Zonation characteristics of the pegmatite (modified from [48]).

Palaeozoic ultrabasic rocks, granites, and pegematite dikes, and the East Tianshan orogenic belt is an important source of gold, silver, copper, nickel, iron, and other metallic minerals in China [see Fig. 1(a)] [43], [44], [45]. Deep and large E–W trending faults, including the Dacaotan fault, Kanggur fault, Yamansu fault, and Aqikekuduke, constitute the basic framework of the East Tianshan orogenic belt, which controls the structure, magmatic events, and mineralization. The Jueluotage belt is an important part of the East Tianshan orogenic belt, which is north of the Middle Tianshan massif, south of the Turpan-Hami (Tu-Ha) basin and Bogega-Halierke belt, and east of Tuokexun close to the Gansu-Xinjiang border. The width of the Jueluotage belt is 30–50 km, and it stretches E–W for more than 600 km [see Fig. 1(a)] [46], [47]. There are early Palaeozoic, Devonian, Carboniferous, Permian, Jurassic, and Quaternary strata in the tectonic belt, and the main strata are carboniferous vocanic-sedimentary strata [48]. The carboniferous strata are composed of three formations, including pillow basalt, spilite, clasolite, and carbonatite in the Wutongwozi Formation; tuff, carbonaceous Si slabstone, siliceous tuff, and siltstone in the Gandun Formation; and tuff, andesitic porphyrite, plagiophyrite, and volcanic breccia in the Xiaorequanzi Formation [49], [50], [51].

The Jingerquan Li–Be–Nb–Ta pegmatite deposit is located in the Jueluotage belt [see Fig. 1(a)]. The outcropping strata are mainly amphibolite of the middle Proterozoic Xianshuiquan Formation and schist of the upper Carboniferous Gandun Formation. The secondary faults (F1 and F2) of the Kanggur

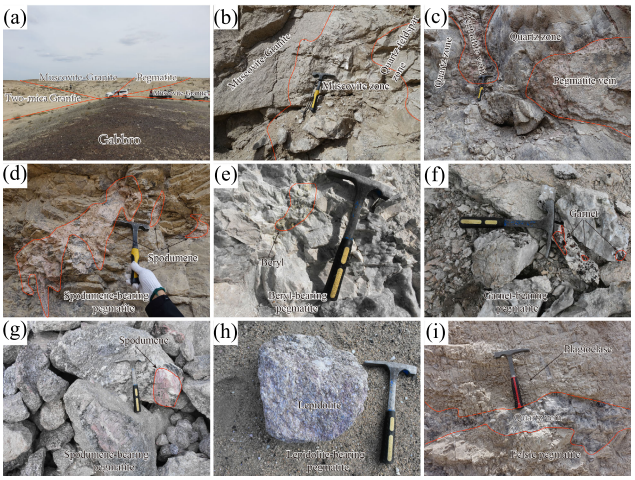


Fig. 2. Photographs of the Jingerquan Li–Be–Nb–Ta pegmatite deposit showing (a) rock mass distribution, (b) zonation of pegmatite, (c) pegmatite veins, (d) and (g) spodumene-bearing pegmatite, (e) beryl-bearing pegmatite, (f) garnet-bearing pegmatite, (h) lepidolite-bearing pegmatite, and (i) felsic pegmatite.

deep fault are observed in the area, which controls the two largest spodumene-bearing pegmatites [see Fig. 1(b)] [53], [54]. The magmatic rocks of biotite granite, muscovite granite, and two-mica granite are widely distributed, with a small amount of gabbroic rocks exposed, which suggests an intrusive contact relationship with the strata [see Fig. 2(a)] [52]. The two ore bodies (No. 1 and No. 2) of pegmatite veins with

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TABLE I
CAMPAIGN SETTINGS FOR THE TWO HYSPEX CAMERAS. A HIGH INTEGRATION TIME WAS USED DUE TO THE HIGH REFLECTANCE TARGETS LIGHT CONDITIONS

Camera	VNIR-1024	SWIR-384
Spectral range (nm)	400-1000	930-2500
Detectors (pixels)	1024	384
Field of view angle (°)	17	16
Extension lens (°)	34	32
Spectral sampling interval (nm)	2.8	5.45
Bands (#)	216	288
Frame (#)	5500	4000

an NE–SW strike distribution occur in muscovite granite [see Fig. 1(b)]. In particular, the No. 1 orebody presents an irregular lenticular shape, which has obvious zonation [see Fig. 1(c)] [48]. The zonation can be divided into a quartz–feldspar zone [see Fig. 2(b)], muscovite zone [see Fig. 2(b)], muscovite granite zone [see Fig. 2(b)], nonmineralized pegmatite zone (quartz–microcline–muscovite), Li-mica-spodumene pegmatite zone (quartz–microcline–spodumene–lepidolite) [see Fig. 2(c), (d), and (g)], and the core of the quartz–spodumene zone from NE–SW based on different mineral assemblages. Be–Nb–Ta mineralization can be seen in the pit outcrop of the No. 2 orebody. The ore minerals are mainly beryl and columbite-tantalite [see Fig. 2(e)], and the gangue minerals are quartz, muscovite, and microcline [see Fig. 2(i)], without Li-bearing minerals. The spodumene-bearing pegmatite displays medium and fine grain sizes and contains major minerals, such as spodumene, quartz, microcline, and lepidolite [see Fig. 2(h)]. Lepidolite is a purple flake aggregate with obvious orientation, and the beryl-bearing pegmatite contains quartz, K-feldspar, and muscovite. Garnet displays coarse grains in the pegmatite [see Fig. 2(f)], with a small amount of muscovite.

III. METHODOLOGY

A. Data Acquisition and Preprocessing

1) Hyperspectral Images and Spectra Collection: Hyperspectral data were acquired using a tripod with a rotation-stage-mounted HySpex hyperspectral imager. The HySpex hyperspectral imager used in this study comprises two sensors (HySpex VNIR-1024/SWIR-384), one covering the visible-to-near-infrared range (VNIR, 400–1000 nm) and the other covering the SWIR range (SWIR, 930–2500 nm), and its performance parameters are shown in Table I. The pit in the study area strikes from the NE to SW, subparallel to the strike of the pegmatite, and the side of the mine has typical zonation characteristics. To obtain a high-quality image, the best recording time was the highest Sun elevation and without a shadow of clouds. Therefore, the data started being recorded at ~1.30 P.M. (the peak position of Sun elevation was ca. 2 P.M. in May). The tripod-mounted sensor faced the strikes of the pegmatite in the pit, viewing the principal vertical mine wall horizontally at a distance of approximately 10 m. Meanwhile, the vertical distance between the rotation platform and the ground was monitored based on the field of view of different sensors. We also carried out hyperspectral recording of the surface ore heap in the mining area. To reduce noise and varying illumination due to Sun azimuth changes and cloud movements, an acquisition window time did not exceed 2 min and recording times at this particular location were determined as per [31]. The parameter combinations of the number of frames and the frame period were recorded, and eight hyperspectral scenes were acquired with each sensor in the same position (i.e., VNIR and SWIR). We selected the scene with the highest spatial resolution and lowest geometric distortion for analysis. For pegmatite with muscovite-granite in the study area, the overall reflectivity of the samples is relatively high. Thus, we chose a Spectralon SRS-99 white reference panel to place in the scene, providing reference spectra for atmospheric correction. The ground-truth reflectance spectra of field and powdered samples were measured in the laboratory using an ASD FieldSpec-4 high-resolution spectrometer. The ASD spectrometer is designed to record 2151 channels throughout a spectral range of 350–2500 nm, with a sampling interval of 1.4–2 nm. The spectral resolution is 3 nm for the 350–1020-nm range and 10 nm for the 1020–2500-nm range. All spectral data were acquired in a darkroom, and light was provided by the quartz–tungsten–halogen lamp of the illuminator. The illumination zenith angle and the azimuth angle were fixed at 30° and 180°, respectively, while the viewing zenith angle and azimuth angle were both fixed at 0°. Ten passes were made for each sample and the data were calibrated using a Spectralon white plate. In this study, we only analyzed the SWIR image scene because most minerals associated with Li pegmatite-related absorption bands occurred within this specific wavelength range.

2) Processing of Outcrop Imaging: In this study, the original acquired RAW hyperspectral image scenes must be accurately preprocessed to identify the specific features of Li-bearing pegmatite minerals. For this, the original image digital number value was first converted to radiation brightness using HySpex RAD software [Norsk Elektro Optikk A/S (NEO), Oslo, Norway]. Second, the bands with low signal-to-noise ratios and affected by water vapor, which has an influence on the reflectance features at 1400 and 1900 nm, were removed. To improve the quality of radiance data, bands 73–85, 159–183, and 280–288 were eliminated, and 241 bands of SWIR were preserved. Finally, atmospheric correction was carried out using ATCOR software developed based on MODTRAN4 radiation conversion by the ReSe Applications Schläpfer Company. Furthermore, a certified reflectance standard white reference panel of known reflectivity was recorded with each image and used as reference material to convert the obtained spectral data into reflectance data. The key to the flat field method is that the radiation value mainly reflects the condition of the solar spectrum obtained in the hyperspectral image, as shown in the following formula:

$$p_{\text{sample}} = \frac{\text{Rad}_{\text{sample}}}{\text{Rad}_{\text{ref}}} \times p_{\text{ref}} \tag{2}$$

$\text{Rad}_{\text{sample}}$ refers to the sample's radiance recorded, Rad_{ref} refers to the reference panel's radiance recorded, and p_{ref} refers to the known reflectance of Spectralon. For hyperspectral image scenes, a region of interest (ROI) of an appropriate

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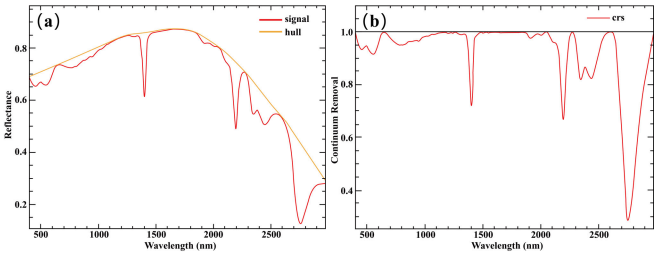


Fig. 3. Continuum-removal technique for reflectance spectra. (a) Continuum-removal (hull quotient) process. (b) Extraction of the absorption features, crs—continuum removed spectra.

white Spectralon panel was selected. The reflectance was calculated by dividing each pixel by the average radiance value of an ROI. The assumed reflectance of the white Spectralon panel is 1; thus, the absolute reflectance of hyperspectral image scenes is obtained by spectral reconstruction after atmospheric correction.

In the following, we select some areas from the obtained outcrop imaging to map pegmatites, mainly considering that the radiation captured by the sensor will be different due to an irregular surface, resulting in some areas with signal distortion in the image. On the other hand, the hyperspectral scenes chosen for analysis were acquired during peak sunlight hours to avoid signal weakening due to the angle of light.

B. Spectral Analysis and Mineral Mapping

1) Extraction of Spectral Features: In the study area, we collected a range of representative rock types, including diorite, biotite-granite, muscovite-granite, two-mica granite, different mineralized dikes, and pegmatite (see Fig. 2). A spectral database was built comprising 120 massive samples and 20 powder samples (block samples were used for thin section identification).

ASD ViewSpec software was used for spectral averaging and spectral correction to eliminate the interference of measurement errors. The spectra collected in the laboratory were normalized by continuum removal (hull quotient), enhancing the spectral absorption features [see Fig. 3(a)]. The continuum removal [55] is the ratio of each point on the actual spectral curve to the value of the corresponding wavelength of the continuum by using the following formulas (3) and (4), calculating the result that maintains the number of wavelengths as the original and in the range of 0–1. IDL DISPEC [56] is used to extract the wavelength position, depth, and other geometric parameters of absorption features [see Fig. 3(b)]

$$\mathrm{CR}_j = \frac{R_j}{R_{\mathrm{start}} + k \cdot \left(\lambda_j - \lambda_{\mathrm{start}}\right)} \tag{3}$$

$$\mathrm{K} = \frac{R_{\mathrm{end}} - R_{\mathrm{start}}}{\lambda_{\mathrm{end}} - \lambda_{\mathrm{start}}} \tag{4}$$

where CR_j is the continuum-removal value of band j, R_j is the original reflectance of band j, K is the slope between the initial point band and the endpoint band in the absorption curve, λ_j is the wavelength of band j R_{start} and R_{end} are the original spectral reflectance of the initial and terminal points in the absorption curve, respectively, and R_{start} and R_{end} are the wavelength of the beginning and ending points in the absorption curve, respectively.

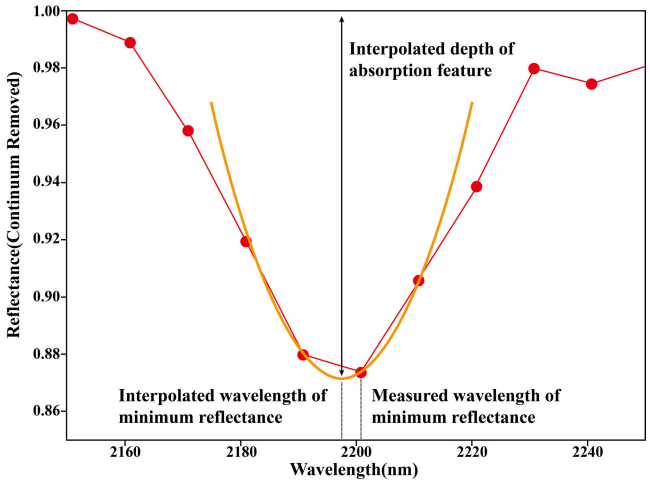


Fig. 4. Interpolation method for determination of the wavelength position of the reflectance minimum of an absorption feature by fitting a parabola through three consecutive measurement points. Fitting was applied to continuum removed reflectance spectra. Wavelength position and depth of the absorption feature were calculated using (4)–(6).

2) Mapping the Wavelength Position of the Deepest Absorption Features: For the hyperspectral image data, the Li-bearing minerals and respective pegmatite zones have specific spectral signatures in the SWIR. The spectral profiles of minerals are affected by a number of factors, such as chemical composition, structure, and surface shape [57]. MWM is an approach that combines the wavelength position and depth information of the deepest absorption features to generate a per-pixel image of proxy for mineralogy and proxy for its mineral abundance [58]. Thus, MWM is a powerful method that uses hyperspectral data and two distinct steps.

Step 1: First, we determined the deepest absorption feature in the SWIR region. The continuum removal is achieved by the convex hull quotient to highlight the position information of the spectral absorption at a specific wavelength, obtaining the depth of the spectral features. Interpolation was used to determine the depth and wavelength position of absorption features, and a second-order polynomial was applied to fit the three consecutive bands where the wavelength position of the lowest reflectance and the corresponding depth of absorption feature was calculated (see Fig. 4) [59]

$$w(\lambda) = a\lambda^2 + b\lambda + c \tag{5}$$

where $w(\lambda)$ is the interpolated reflectance value at position λ; λ is the wavelength position in nm; and a, b, and c are the coefficients of the parabola function.

$$w_{\min} = -\frac{b}{2a} \tag{6}$$

where $w_{\min}$ is the interpolated wavelength position at minimum reflectance and a and b are the coefficients of the parabola function.

$$\mathrm{depth} = 1 - \mathrm{f}(w_{\min}) \tag{7}$$

where depth is the interpolated depth of absorption feature.

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The output result of step 1 contained two parameters: 1) the interpolated wavelength of minimum reflectance and 2) the interpolated depth of the absorption feature. A spectral subset of step 1 would have determined the spectral range of the deepest features. The step 1 subset is set in the 2000–2500-nm spectral range in which many minerals of interest for this study show diagnostic absorption features.

Step 2: The determined wavelength position information and the depth information are fused into an RGB color image, and then, an RGB color image is transformed into HSV color space, where the wavelength position of the absorption features controls the hue, whereas the intensity of the color image is dependent on the depth of the absorption features. In this study, 40% was selected as the depth stretch value to ensure good intensity contrast and avoid displaying noisy and indistinct features. We selected the wavelength range of the spectral subset of step 2 from 2100 to 2450 nm, which was a subset of step 1, focusing on the position and the deepest absorption feature.

In the final resulting maps, the per-pixel overview maps produced by MWM are not classified mineral maps that require a geologist to interpret the main spectral characteristics of minerals. Its advantage is that users can combine spectral features to map the spatial pattern of the position and depth information without prior knowledge of the area and formal spectroscopy training. At the same time, in order to facilitate the follow-up field verification and sample collection work, SAM and spectral information divergence (SID) are used to draw the mineral distribution map of the ore heap. Compared with MWM, classification methods, such as SAM and SID, can provide an indirect tool to identify either the target minerals and similar spectral features of proxies. According to the collected sample spectra, the spectra of Li-mica-spodumene pegmatite are used as the reference spectra for both SAM and SID methods.

IV. Results

A. Identification of Spectral Features of Relevant Pegmatites on Hand Samples

For comparison with the map panorama of the wavelength position and the deepest absorption features by MWM, the spectral characteristics of spodumene-bearing pegmatite, lepidolite-bearing pegmatite, Li-mica-spodumene pegmatite, nonmineralized pegmatite, and muscovite-granite are shown in Fig. 5. The main mineral of spodumene-bearing pegmatite is spodumene, with a small amount of quartz and feldspar. The main mineral of lepidolite-bearing pegmatite is lepidolite, with a small amount of quartz and feldspar. The main minerals of Li-mica-spodumene pegmatite are spodumene, lepidolite, quartz, feldspar, and other minerals. We selected these samples because they represent the mineralogical zonation and the various degrees of Li-mineralization of pegmatite, i.e., Li-rich pegmatites and Li-poor pegmatites. To analyze spectral characteristics accurately and provide a theoretical basis for MWM, each hand sample contains three types of spectra, including the original spectrum, continuum-removal spectrum, and spectral curve in the range of 2000–2500 nm, which were collected from the fresh block sample, weathered block sample, and powder sample (see Fig. 5).

1) Identification of Li-Rich Pegmatite Spectral Features: In this article, spodumene-bearing pegmatite is one of the main Li-rich pegmatites in the study area. It exists as pink crystals with vitreous luster. The spodumene at the edge of quartz has a large crystal form, and it also coexists with feldspar, lepidolite, and muscovite. The spectra of spodumene-bearing pegmatite were acquired considering pure and complete crystal spodumene intergrowth with muscovite and quartz [see Fig. 5(A)]. Fig. 5(a) shows that the reflectance of spodumene-bearing pegmatite increases rapidly in the region of ∼350 to ∼680 nm. After that, it decreased first and then increased from ∼680 to ∼1300 nm. The overall reflectance remained high from ∼1400 to ∼1850 nm. As shown in Fig. 5(b), the main absorption characteristics are located at ∼1410, 1910, ∼2210, ∼2345, and ∼2445 nm, the asymmetric features at ∼1410 and ∼1910 nm are related to hydroxyl and molecular water, and there is a slight inflection at approximately 1460 nm, indicating the existence of montmorillonite in the samples. The noticeable asymmetrical absorption feature of Al–OH at approximately 2200 nm indicates the possible existence of kaolinite. However, spodumene is a clinopyroxene group mineral that has no diagnostic spectral features in the VNIR and SWIR bands but only in the thermal infrared wavelength range [60]. Therefore, the spectral features of spodumene-bearing pegmatite in the SWIR region may be caused by clay alteration [61], [62]. Meanwhile, the features' characteristic of absorption at approximately 535 and 965 nm is related to iron incorporation (Fe^{3+} and Fe^{2+}). Fig. 5(c) shows that the absorption depth of clay alteration at ∼2210 nm is approximately 0.3, and the absorption depth corresponding to other absorption features is approximately 0.1.

Fig. 5(B) shows that lepidolite-bearing pegmatite presents coarse and medium purple aggregates, which are the main phase of the quartz–feldspar–lepidolite assemblage. Second, lepidolite with small flakes also coexists with spodumene, muscovite, and other minerals. The spectral characteristics of lepidolite-bearing pegmatite are similar to those of the spodumene-bearing sample, and lepidolite has an SWIR spectral signature similar to white micas. The reflectance increases slowly from 350 nm and has the same trend at ∼680 to ∼1400 nm. Compared with the reflectance spectra of the spodumene-bearing sample, it maintains a high reflectivity over a long spectral range (between 1450 and 2050 nm) [see Fig. 5(d)]. Lepidolite-bearing pegmatite has more absorption features than the spodumene-bearing sample and inherits the typical absorption characteristics of muscovite. The narrow-thin absorption trough at ∼1410 nm and the wide-shallow absorption trough are associated with hydroxide ions and water molecules. The strong absorption features located at ∼2197, ∼2344, and ∼2430 nm are interpreted as the main Al–OH absorption and related secondary absorption [63], [64]. The absorption features at ∼550, ∼960, ∼1248, and ∼1326 nm are interpreted to be related to ferric and/or ferrous iron [see Fig. 5(e)]. Fig. 5(f) shows that the absorption depth of approximately 2200 nm of lepidolite-bearing pegmatite is over 0.6, and the rest are less than 0.5.

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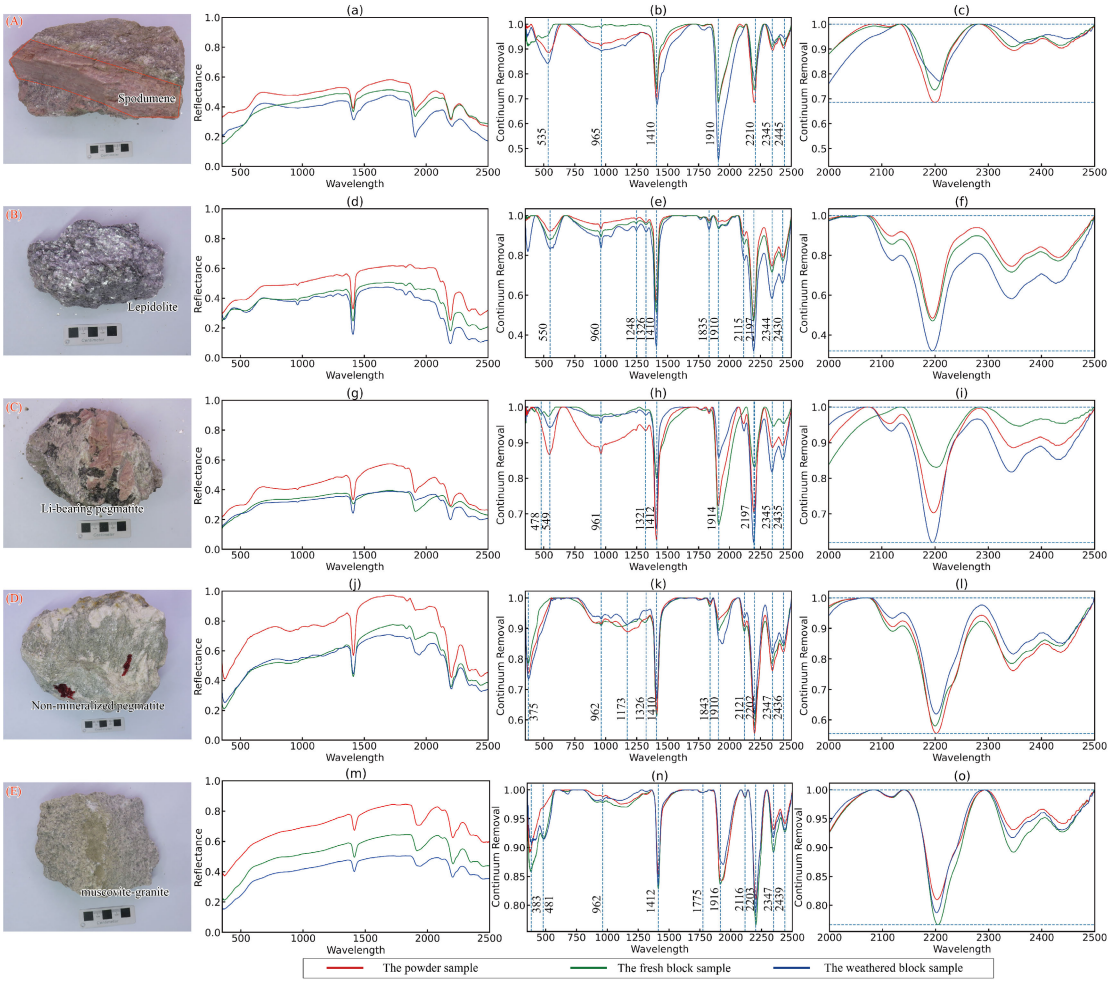


Fig. 5. Spectra of pegmatites from the hand samples captured by the ASD. (A)–(E) Hand samples of spodumene-bearing pegmatite, lepidolite-bearing pegmatite, Li-mica-spodumene pegmatite, nonmineralized pegmatite, and muscovite granite. (a)–(c) Spodumene-bearing pegmatite spectrum. (d)–(f) Lepidolite-bearing pegmatite spectrum. (g)–(i) Li-mica-spodumene pegmatite spectrum. (j)–(l) Nonmineralized pegmatite spectrum. (m)–(o) Muscovite granite spectrum.

The main minerals of Li-mica-spodumene pegmatite are spodumene, lepidolite, quartz, feldspar, and a small amount of muscovite [see Fig. 5(C)]. Therefore, the spectral characteristics of ore-bearing pegmatite are similar to those of spodumene-bearing pegmatite and lepidolite-bearing pegmatite, and the reflectance of the powder samples is higher than that of the fresh and weathered block samples [see Fig. 5(g)]. Fig. 5(h) shows that ore-bearing pegmatite has absorption features of both spodumene-bearing pegmatite and lepidolite-bearing pegmatite at positions of ∼478, ∼549, ∼961, ∼1321, ∼1412, ∼1914, ∼2197, ∼2345, and ∼2445 nm. In the region of ∼2000 to ∼2500 nm, each depth of absorption is within 0–0.5 [see Fig. 5(i)], and the spectrum is more similar to lepidolite-bearing pegmatite, indicating that lepidolite-bearing pegmatite occupies a large proportion in the measured field of view.

2) Identification of Li-Poor Pegmatite Spectral Features: The Li-poor pegmatite was sampled between the graphic texture of the quartz and feldspar zone and the muscovite granite zone, presenting green cluster-like aggregates, which may also coexist in flakes with quartz–feldspar–spodumene assemblage [see Fig. 5(D)]. The overall reflectance of Li-poor pegmatite is higher [see Fig. 5(j)]. Similarly, the absorption features located at ∼1326 and ∼1410 nm are overtones of the hydroxyl-related fundamental stretching vibrations [65]. The ∼1843-nm absorption is related to H_2O, and the absorption features at ∼2121, ∼2202, ∼2347, and ∼2436 nm are related to the combination of the stretching and bending vibrations of the hydroxyl group. The remaining absorption features at ∼375, ∼961, and ∼1173 nm are related to Fe^{2+} and Fe^{3+} [see Fig. 5(k)]. Notably, the full-width at half-maximum (FWHM) corresponding to the absorption features at a position of 2200 nm was concentrated at approximately 60 nm, and the asymmetry of the Li-poor pegmatite was significantly higher than that of the lepidolite-bearing pegmatite [see Fig. 5(l)].

The fine-grained muscovite granite is mainly composed of quartz, feldspar, and muscovite and is the wall rock of Li-bearing pegmatite [see Fig. 5(E)]. The spectral characteristics of muscovite granite inherit the spectral characteristics of muscovite [see Fig. 5(m)]. Muscovite granite has obvious absorption features at ∼1400, ∼1900, and ∼2200 nm because quartz and feldspar have no characteristic absorption in the

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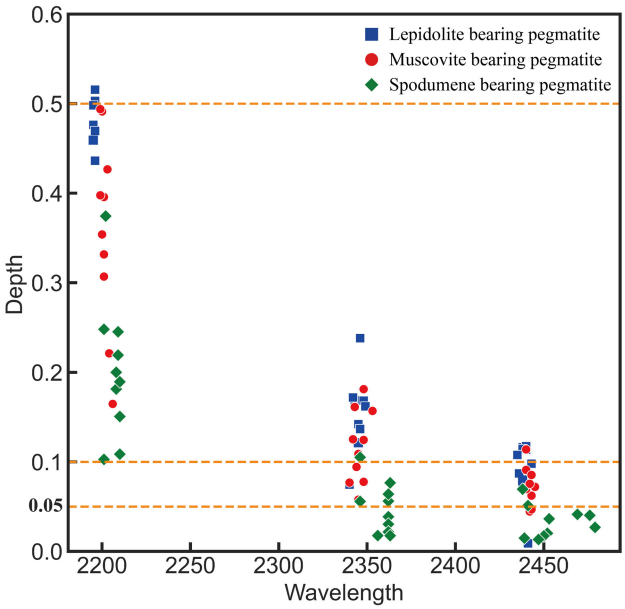


Fig. 6. Position and depth of information for the absorption features in the SWIR.

region of ∼350 to ∼2500 nm, leading to the highlight of the spectral behavior of Al–OH minerals [see Fig. 5(n)]. The corresponding absorption depths of muscovite granite at ∼2200, ∼2347, and ∼2439 nm are smaller than those in the muscovite sample [see Fig. 5(o)], which may be determined by the percentage of muscovite in the measurement field of view.

B. Wavelength Position Maps

In this study, ten samples of each pegmatite, including lepidolite-bearing pegmatite, muscovite-bearing pegmatite, and spodumene-bearing pegmatite, were used for statistical analyses of the wavelength position and depth information in the SWIR region (see Fig. 6). The diagram shows that the absorption depth around ∼2200 nm of most samples is approximately 10%–50%. Specifically, compared with Li-poor pegmatite represented by muscovite-bearing pegmatite, the shift of the absorption position at ∼2200 nm to the shortwave direction of Li-rich pegmatite represented by lepidolite-bearing pegmatite has been observed, which may be related to the change in Al ion content caused by the exchange of Li ions [15], [65]. The characteristic absorption position at ∼2200-nm peaks of spodumene-bearing pegmatite was located at ∼2210 nm. The variation range of the absorption depth around ∼2350 nm was between 5% and 20%, indicating that the absorption characteristics were not obvious. The variation range of the position at ∼2450 nm was between 2430 and 2490 nm, and the absorption depth was less than 10%, which makes it difficult to distinguish Li-rich pegmatite from Li-poor pegmatite. Therefore, based on the above statistical results, each hyperspectral image was mapped with MWM to produce two per-pixel overview maps combining the position and depth information of the deepest absorption feature. The region of ∼2000 to ∼2500 nm is selected as the first subset of the step to calculate the position of absorption features caused by the combination of hydroxyl and cations, avoiding the interference of the absorption features of ∼1400 and ∼1900 nm caused by water molecules and hydroxyl, and other weak absorption features. Step subset 2 sets the region smaller than ∼2000 to ∼2500 nm to show more details in the absorption feature wavelengths of the individual groups. We adjusted the depth stretch of step 2 to eliminate interference from spectral noise (see Table II).

TABLE II
MINIMUM WAVELENGTH PRODUCTS PRODUCED AND THE PARAMETERS USED

	SWIR overview	SWIR 2150-2250
Step 1 Wavelength of Minimum	2100nm to 2450nm	2150 nm to 2250nm
Step 2 Wavelength Mapping	2100nm to 2450nm	2150nm to 2250nm
Step 2 depth stretch	0% - 50%	10%-50%

1) Regional View of Pit: Fig. 7 shows the deepest absorption feature wavelength position and depth image of hyperspectral images between ∼2100 and ∼2450 nm. Due to the limitation of light conditions, three subimages on the side of the pit were selected [see Fig. 7(a)]. Fig. 7(a) shows an image obtained from the VNIR, which is used as a verification for MWM to extract useful ore-bearing information from the SWIR images. Fig. 7(1-b) shows a large area of green pixels with a wavelength of ∼2200 nm, which indicated Al–OH minerals. A small number of yellow areas appear, which are caused by an alteration mineral dominated by chlorite in pegmatite. In Fig. 7(1-c), the pixel distribution map in the region of ∼2150 to ∼2250 nm is shown. Most of the yellow-green regions are ∼2200 to ∼2210 nm, which corresponds to the absorption peak of Al–OH minerals. A shift of the position at ∼2200 nm to the longwave direction of Al–OH minerals has been observed, and it is possible that muscovite is located in the graphic texture of the quartz and feldspar-muscovite zones [see Figs. 1(c) and 2(b)]. Two areas of yellow pixels with absorption features near 2215 nm and red pixels with a wavelength position of approximately 2220 nm are mapped, which is due to the Tschermak Exchange (tetrahedrally and octahedrally coordinated Al^{3+} being replaced by Si^{4+} and divalent cations such as Fe^{2+} and Mg^{2+}) [66]. In Fig. 7(2-b), a large number of pixels are in the range from 2350 to ∼2450 nm, and the absorption characteristics were not obvious, which corresponded to a lower amount of Al–OH minerals in Fig. 7(a-2). The area of dark-green pixels with a wavelength position at approximately 2200 nm is scattered in irregular strips, corresponding to the vein-like spodumene of pegmatite in Fig. 7(a-2). The pixels corresponding to wavelengths of ∼2400 to ∼2450 nm are distributed throughout the subimage. From the shallower features at these wavelengths of the pixels and spodumene-bearing pegmatite, it is complicated to interpret the spectra and the differentiation of mineralogical signatures from noise. Fig. 7(2-c) shows the area distribution map of pixels with a variation range of ∼2205 to ∼2215 nm, which is consistent with the green area in Fig. 7(2-b), which corresponds to the Al–OH of altered minerals. The pegmatite deposit has the characteristics of zoning, from northeast to southwest, including quartz–spodumene zone, ore-bearing pegmatite zone (mainly spodumene–microplagioclase–muscovite

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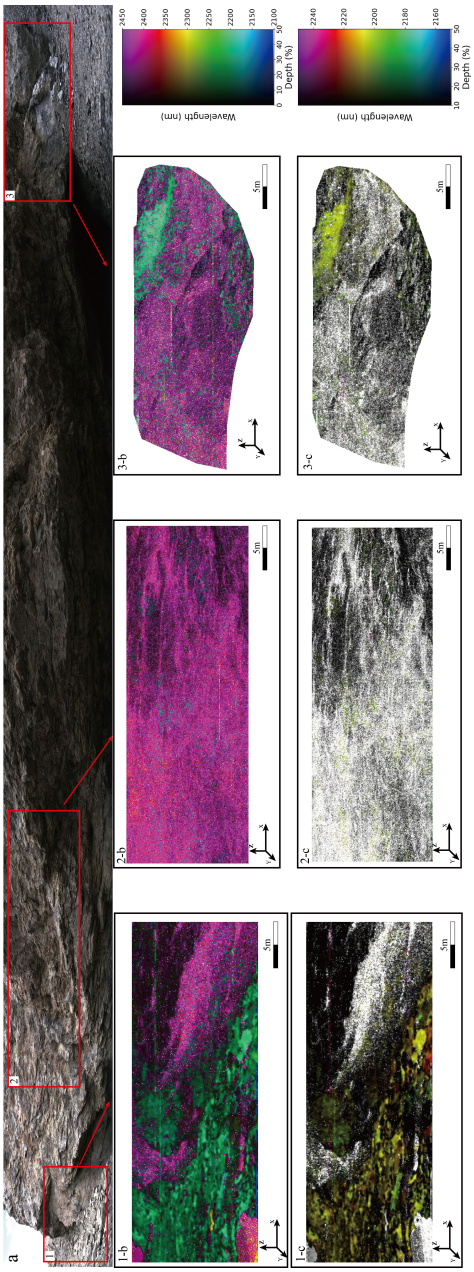


Fig. 7. Overview map of the SWIR wavelength mapping results in the south side of pit. The hues represent the wavelength of the dominant spectral features in the SWIR range (see the legend for the link with wavelength). The color intensity represents the feature depth and is a proxy for abundance or importance within the pixel. (a) VNIR image. (b) Wavelength position maps at 2100–2450 nm. (c) Wavelength position maps at 2150–2250 nm.

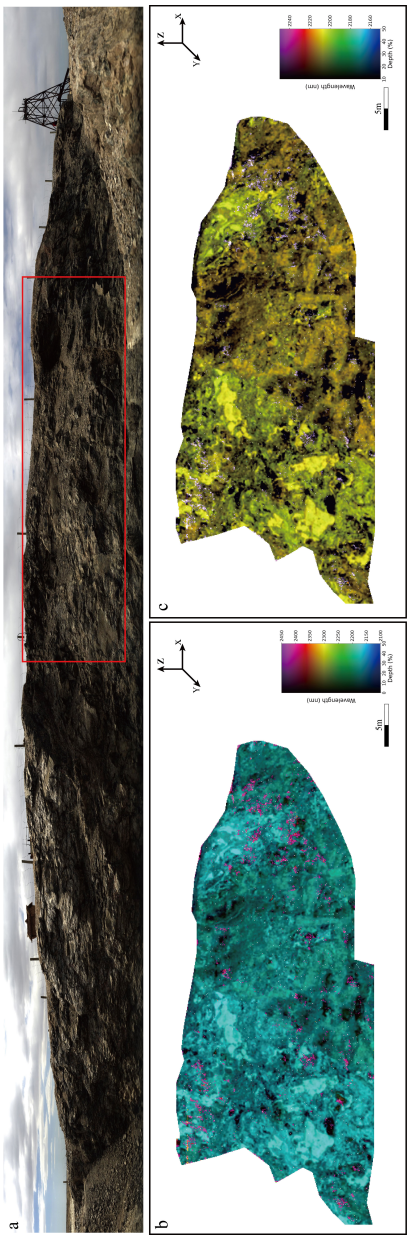


Fig. 8. Overview map of the SWIR wavelength mapping results in the north side of pit. (a) VNIR image. (b) Wavelength position maps at 2100–2450 nm. (c) Wavelength position maps at 2150–2250 nm.

zone), quartz–microplagioclase–muscovite zone, graphic texture zone, muscovite granite zone, muscovite zone, and graphic texture zone Fig. 7(a-1)–(a-3). The light-green pixel area in Fig. 7(3-b) corresponding to the muscovite zone is in a wavelength position at ∼2200 nm. Bright yellow-green pixels indicate the Al–OH absorption features at a wavelength near 2209 nm. The black pixels show the absence of absorption features between ∼2100 and ∼2450 nm and represent the quartz zone. The dark green pixels related to lepidolite-bearing pegmatite or ore-bearing pegmatite show Al–OH absorption features near 2198 nm in Fig. 7(3-c).

The pegmatite veins on the other side of the pit are in a disorderly distribution, which leads to confusion in the zoning characteristics of the pegmatite, so a small area is intercepted for analysis [see Fig. 8(a)]. The dark green pixels with relatively long wavelength positions occur between two clusters of cyan-colored pixels with relatively shorter wavelength positions [see Fig. 8(b)]. Minimum wavelength maps that were calculated over the wavelength ranges from ∼2190 to ∼2210 nm show the layered silicate mineral absorption features that are likely related to the content of Al–OH. The black pixels with no absorption features between ∼2150 and ∼2250 nm occur in the nonore-bearing pegmatite zone (mainly composed of quartz, feldspar, and low-abundance muscovite), and the distribution of black pixel areas in the entire image indicates that multiple pegmatite veins exist on the side. Bright

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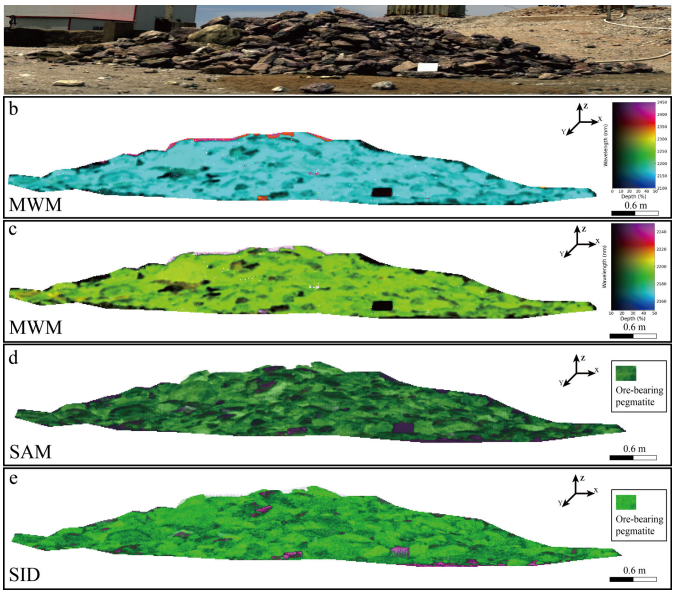


Fig. 9. Overview map of the SWIR wavelength mapping results in the ore heap. (a) VNIR image. (b) Wavelength position maps at 2100–2450 nm. (c) Wavelength position maps at 2150–2250 nm. (d) SAM result. (e) SID result.

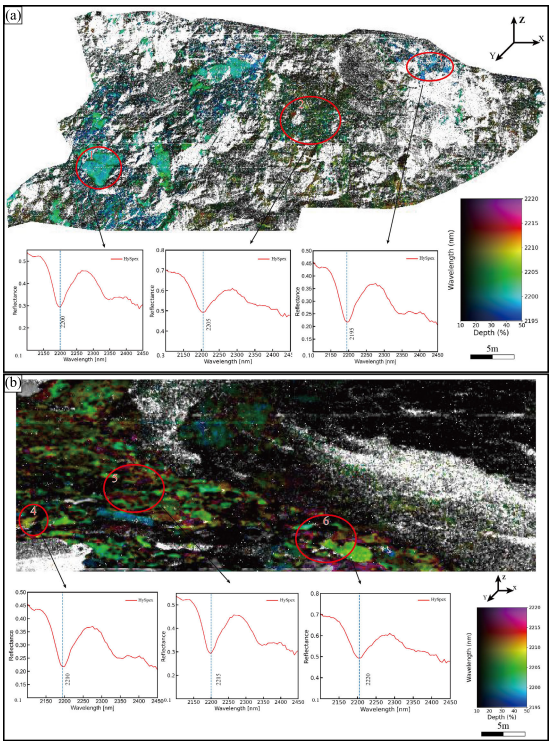


Fig. 10. (a) Overview map of the 2195–2220-nm wavelength mapping results in the north side of pit. Average reflectance spectra of ROIs, 1–3. (b) Overview map of the 2195–2220-nm wavelength mapping results in the south side of pit. Average reflectance spectra of ROIs, 4–6.

yellow pixels indicate Al–OH absorption features of intermediate Al-content minerals at a wavelength near ∼2209 nm, that is, the Al–OH absorption position tends to the longwave direction, which may be caused by the decrease in Al^{VI} content and the increase in Si^{4+} in Al–OH minerals, where the alteration products in plagioclase are Al–OH minerals. The yellow-green pixels show the Al–OH absorption features near ∼2202 nm in the nonmineralized pegmatite zone (quartz–microplagioclase–muscovite). The dark yellow pixels indicate Al-poor minerals, which are in the middle of the nonore-bearing pegmatite [see Fig. 8(c)].

2) Ore-Bearing Pegmatite Heap: The pegmatite heap includes Li-mica-spodumene pegmatite, spodumene-bearing pegmatite, and lepidolite-bearing pegmatite, which were selected due to their various Li contents. Fig. 9(b) shows that most of the material in the ore heap is characterized by strong absorption at approximately 2200 nm. Most of the area indicates a wavelength position of ∼2175 to ∼2200 nm, which is typical of Al–OH containing pegmatites, such as lepidolite-bearing pegmatite and Li-mica-spodumene pegmatite. A small part of the area corresponds to the wavelength position between ∼2200 and ∼2210 nm, which is typical for intermediate to Al-poor white micas but also kaolinite [see Fig. 9(b)]. SID and SAM results [see Fig. 9(d) and (e)] show that pixels in green are ore-bearing pegmatite, corresponding to the pixels of the MWM result. On the other hand, SID and SAM results indicate parts that spectroscopically similar to host rocks containing minerals, such as lepidolite and muscovite. It is difficult for pixels to visually display the changes in spectral characteristics caused by pegmatite zonation. The combination of all three methods provides us with possible indications of Li-rich pegmatite.

Based on the results of previous studies [24], [39], [62], the characteristic ranges of stable absorption of pegmatite samples are approximately 1410, 1910, 2200, 2350, and 2450 nm (see Fig. 5). The Al–OH containing mineral map with a wavelength position near 2200 nm was produced by MWM to assign distinct colors in the region of ∼2150 to ∼2250 nm to distinguish layered silicate minerals with different Al contents [see Fig. 9(c)]. Most of the yellow-green pixel wavelength positions appear near ∼2198 nm, which are likely related to the area of lepidolite-bearing pegmatite. The green pixels surrounding the yellow-green pixel area in a wavelength position of absorption features at ∼2210 nm are associated with Mg–Fe-rich phengite in Li-poor pegmatite.

The map in Fig. 10(a) combines the wavelength position of the deepest features and their depths from ∼2195 to ∼2220 nm to evaluate the validity of the pixel map by MWM. Bright cyan-green pixels [see Fig. 10(a)] of Li-rich pegmatite indicate that the absorption feature at ∼2200 nm shifts to shorter wavelengths. Most of the pixels show absorption positions in the region of ∼2195 to ∼2205 nm, which may be related to the Li content in Li-rich pegmatite. Fig. 10(b) shows that the bright green pixels occupy the main part because the muscovite zone has an obvious absorption feature at ∼2205 nm. Meanwhile, there are pixel distribution maps corresponding to the obvious absorption feature of ∼2210 to ∼2220 nm in the muscovite zone.

C. Validation With Thin Sections and Spectra of Hand Samples

To validate our results, spectra from the hyperspectral images (see Fig. 10) were compared with spectra taken by the handheld spectroradiometer of each hand samples (see Fig. 5). Spectra from the hyperspectral images show characteristic pegmatite zones' absorption features at ∼2200 nm (see

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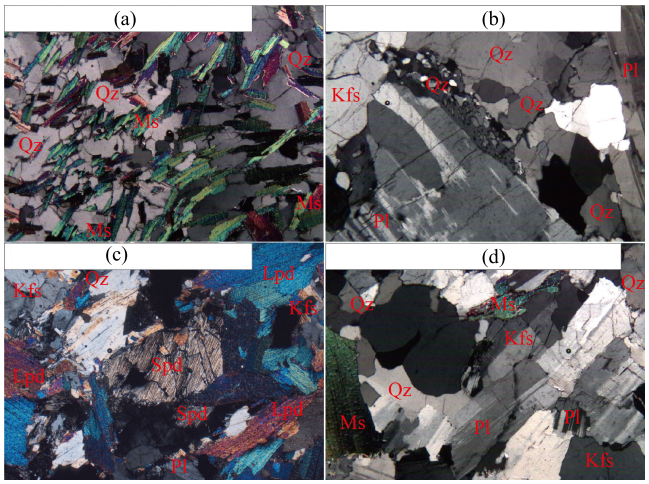


Fig. 11. Thin sections of different types of pegmatite mineralogy. (a) Quartz–feldspar zone. (b) Nonmineralized pegmatite zone. (c) Li-mica-spodumene pegmatite zone. (d) Muscovite granite zone. Qz: quartz. Ms: muscovite. Kfs: K-feldspar. Pl: plagioclase. Spd: spodumene. Lpd: lepidolite.

Fig. 10), and the distinguishing characteristic of this feature is the presence of an offset at 2200 nm. On the other hand, we carried out comprehensive geological field work in the study area, especially detailed sampling of two ore bodies (No. 1 and No. 2) of pegmatite veins. Using thin sections, we identified mineral assemblages in hand specimens that are typical for the quartz–feldspar zone, mineralized pegmatite, pegmatite, and muscovite granite, as shown in Fig. 11.

The quartz–feldspar zone is mainly composed of plagioclase and quartz, with a small amount of muscovite. The content of plagioclase with hypidiomorphic granularity is 60–65 vol%, and the quartz content is 25–30 vol%, often symbiotic graphic texture with plagioclase [see Fig. 11(a)]. Nonmineralized pegmatite displays massive structure. The main minerals are plagioclase, quartz, and muscovite, and the secondary minerals are sericite and opaque minerals. The content of plagioclase with hypidiomorphic tabular texture (2)–10 mm in size) is 65–70 vol%. The content of quartz with hypidiomorphic granular texture is 30–35 vol%. Muscovite with lamellar texture is distributed randomly [see Fig. 11(b)].

Li-mica-spodumene pegmatite reveals a massive structure, the main minerals are plagioclase and quartz, and the secondary minerals are lepidolite and spodumene. The content of plagioclase with hypidiomorphic granularity is approximately around 60 vol%. The content of lepidolite with lamellar texture is 17 vol%, which is purple and blue under a microscope. It appears as aggregates. The content of spodumene panidiomorphic-hypidiomorphic granular texture is 10 vol% [see Fig. 11(c)]. Muscovite granite displays medium and fine grain sizes (0.5–3 mm). The main minerals are plagioclase, quartz, and muscovite, and the secondary minerals are lepidolite and garnet [see Fig. 11(d)]. The content of quartz with allotriomorphic granular texture is 20–25 vol%. The content of plagioclase with hypidiomorphic granular texture is 70–75 vol%. Garnet has a hypidiomorphic granular texture.

V. Discussion

Hyperspectral images were collected from the different pegmatite zones in the open pit and processed using the MWM method. Each color hue is associated with a particular wavelength of the main spectral feature, resulting from the SWIR hyperspectral dominant mineralogy in the respective pixel. Meanwhile, the feature depth is interpreted as a proxy for relative mineral abundance and alteration in the pegmatite zone. The step 1 range of 2100–2450 nm was chosen to include the SWIR spectral feature of pegmatite and alteration mineral feature. The range of the 2150–2250-nm overview map in step 2 exhibits slightly different hues, but these features more accurately describe the spectral characteristics of pegmatite deposit mineralization.

Pegmatites often exhibit zoning due to different mineral assemblages and chemistry [67], [68], [69]. Al–OH minerals occur in all zones of these Li-rich pegmatite, Li-poor pegmatite and alteration minerals, corresponding to the zones of muscovite, ore-bearing pegmatite, and so on, and alteration minerals of cookeite and clays (such as illite, montmorillonite, and kaolinite). The mineral chemistry of white mica and other SWIR active mineral species can be dependent on a large range of factors, including pressure, temperature, and redox conditions [66], [70], [71], [72]. The Al–OH absorption position at 2200 nm of Li-rich pegmatite associated with albite shifts to the left (<2200 nm), and that of Li-poor pegmatite associated with potassium feldspar shifts to the right (>2200 nm). This also indicates the change in metallogenic environment from Al-rich to Si-rich. The hyperspectral imaging obtained in main ore body No. 1 shows that the Li-bearing pegmatite has an irregular lenticular shape [48]. The ore minerals of spodumene, lepidolite, and beryl are evident in the mining area, while the gangue minerals of quartz, albite, microcline, and muscovite are exposed in various zones of Li-poor pegmatite. The quartz–spodumene zone is formed by magmatic crystallization in the early stage, which does not have absorption characteristics in the VNIR-SWIR. The dark green pixels at 2215 nm in Fig. 7(3-b) may correspond to cookeite. The limitations of the outcrop size of spodumene and outcrop sensing result in mixed spectral signatures. Therefore, the mineral surface maps show relative abundances of mica series minerals mixed with quartz and feldspar.

The intergrowth of spodumene and lepidolite with small crystals is formed by late hydrothermal metasomatism. From two-mica granite-muscovite granite-pegmatite, the content of quartz in the rock-forming minerals does not change significantly, but the content of purple lepidolite increases obviously. The common alteration of surrounding rock is muscovitization, albitization, and so on. Among them, Li is mainly incorporated into micas in three ways in the evolution of the muscovite–lepidolite series' composition [72], [73], [74]: $2Si^{IV} + Li^{VI} \leftrightarrow 3Al^{Tot}$, $3Li^{VI} \leftrightarrow Al^{VI} + 2\square^{VI}$ and $4Al^{Tot} \leftrightarrow 3Si^{IV} + 2\square^{VI}$. It is assumed that Li was a substitute for Al^{VI} in octahedral sites and the replacement reaction of silicon and aluminum [75]. Thus, the absorption wavelength of Al–OH in Li-rich pegmatite tends to be in the shortwave direction, while the absorption wavelength of Al–OH in Li-poor pegmatite tends to be in the longwave direction [62], [76].

We are able to quantitatively describe Jingerquan pegmatite zones and identify the area of Li-rich and Li-poor pegmatite,

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but maps of minerals represented by molecular groups (Al–OH bearing minerals) produced are purely surface estimates and cannot directly serve as quantitative concentrations of minerals and pegmatites. In terms of rapid exploration of pegmatites, it is essential to conduct semiquantitative analysis of pegmatite zones and mineralized areas on a macroscale, document the spectral characteristics of wavelength positions and depths in the different mineral assemblages, and generate the optimal pegmatite zone classification and metallogenic environment [77]. We also produced two mineral maps (see Fig. 10) that correspond to relatively complete and obvious walls of the Jingerquan pegmatite zones, in which the spatial distribution of Al–OH minerals is discernible. The absorption position of Al–OH in the SWIR region from ∼2195 to ∼2220 nm is an important indicator of the characteristics of the pegmatite zone (Li-poor pegmatite to Li-rich pegmatite series). Among them, the main spectral absorption characteristics of feldspar (albite, potassium feldspar, and so on.) are in the longwave infrared (LWIR) region, and feldspar exists as large crystal and graphic crystal textures in pegmatite zones. In Figs. 8–10, it can be seen that they are extensively replaced by other minerals, and the secondary minerals are mainly Al–OH minerals [78], [79], [80] [81], [82]. With hyperspectral imaging, we were able to determine the degree of Li-mineralization and alteration associated with mineral assemblages that may indicate the physical and chemical properties of the ore-forming material.

In many cases, heterogeneous minerals with large crystal sizes and the complexity of alteration and evolution series of minerals in pegmatite zones make the hyperspectral imaging approach unique. Taking a hyperspectral imaging approach allowed us to determine a wide range of zoning characteristics of Li-rich pegmatite, Li-poor pegmatite, and alteration to highlight potential metallogenic spaces. However, a challenge in correctly identifying minerals in pegmatites may come from several aspects, including the spectral confusion with Li-bearing minerals (Li-mica-lepidolite series) and alteration minerals (cookeite-clay series) in SWIR data, the fusion of the size of pegmatite outcropping and the resolution of hyperspectral images, and spodumene, and many other Li-bearing minerals produce any absorption features in the SWIR. In the future, it will be necessary to establish higher spectral and spatial resolutions and wider spectral range image data of Li-minerals to serve prospecting using remote sensing from multiscale and multisource data.

VI. Conclusion

In this article, we demonstrated the ability of MWM to map outcrop pegmatite zones from HySpex imaging hyperspectrometer SWIR data. Wavelength position images of the pit and ore heap in the Jingerquan Li–Be–Nb–Ta pegmatite deposit were calculated to provide a regional overview of groups of minerals with spectral absorption features. The results showed that images of the wavelength position and the depth of the deepest absorption features between ∼2100 and ∼2450 nm combine the information of the dominant spectral feature in each pixel for all the different mineral assemblages that display strong spectral features. In particular, the image of wavelength position and the depth of deepest absorption features between ∼2195 and ∼2220 nm provide an efficient assessment of Li-rich pegmatite, Li-poor pegmatite, and alteration, such as mineral abundance and series transformation of minerals. Different pegmatite zones show a wavelength shift of the Al–OH feature, which is probably due to mica species chemistry with respect to the octahedral cation composition (i.e., Al versus Li versus Fe).

Compared with SAM and SID methods, we conclude that MWM provides a useful method for the exploration of Li-rich pegmatite directly and rapidly using hyperspectral images, and the Al–OH feature is used as one of the indicators to discriminate parts of the orebody and host rocks in the Jingerquan mining area. The absorption position of Al–OH in Li-rich pegmatite is located in the shortwave direction (2195–2205 nm), while those in Li-poor pegmatite are located in the longwave direction (2208–2220 nm). The results of the exploration analysis can be used as the spectroscopic basis for inferring the surface mineralogy of Li-bearing assemblages. The next step would be to establish a laboratory imaging hyperspectral database of pure Li-bearing minerals and non-Li-bearing minerals so that hyperspectral images of Li-bearing pegmatites can be analyzed at a multiscale approach to map minerals and their spatial distributions.

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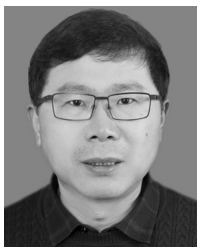
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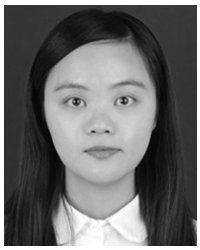
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